

Ian Ronk

Head of Data · Engineer & Researcher

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Production data systems and the data engineering behind research: Airflow, DuckDB and PostGIS pipelines that turn open spatial data into papers. Leads a team of 4 as Head of Data; interest in research on urban dynamics.

TOOLS

Python · SQL · PostGIS · Airflow · DuckDB · PyTorch · MLflow · Bash · LLM pipelines

PROFESSIONAL EXPERIENCE

Head of Data, KR&A, Amsterdam

Jul 2025 – present

- Lead a team of 4 through a transformation of the product offering, serving pension funds and leading FinTechs.
- Develop, maintain and expand data pipelines (e.g. a weekly scrape of 300k records) and spatial big data products.
- Delivered a global connectivity score: 1TB+ processed across 13 servers into a production API.
- Represent the data function with clients: defending methodology against PhD-level scrutiny, presenting to portfolio managers and senior stakeholders.
- Drive AI adoption: OCR and LLM document extraction, agentic pipeline monitoring, agent-assisted development.

Independent Researcher, urban dynamics

2025 – present

- Self-directed research programme: two working papers and a method paper in preparation (see Research), each backed by an open, reproducible pipeline.

Medior Data Scientist, KR&A

Jun 2022 – Jul 2025

- Project lead for two multi-year projects, including a 3-year hedonic house-price-index project for Eurostat: scraping, storing and managing the data, building the regressions, interpreting results.
- Restructured data infrastructure from legacy systems to Airflow, Iceberg and FastAPI.
- Client-facing throughout, with CBS, Eurostat and pension funds.

Junior Data Scientist, KR&A

Oct 2021 – Jun 2022

- Flood-occurrence prediction from alternative data (BSc-thesis project); 90%+ accuracy in risk classification.
- Improved API efficiency tenfold through spatial optimisations, resulting in promotion.

Junior Full-Stack Developer, Exact (former SRXP), part-time

Sep 2019 – Sep 2022

- Enterprise expense-declaration software (EmberJS, PHP) under CI/CD and testing.
- Maintained a client-facing webapp, working on business logic and styling.

RESEARCH TOPICS

- **Calibrating Free Postcode Boundaries from OpenStreetMap**, Release August 2026. Seed-density-to-LoU calibration of an OSM-Voronoi pipeline; NL/DK references, transfer to BE, applied to Italy's 4,209 CAP polygons.
- **When Does Metro Infrastructure Capitalize into Property Prices?**, Release September 2026. Phase-decomposed DiD across seven European cities, $n = 42,004$; wild-cluster-bootstrap inference. Found some correlation, but not strong.
- **Building an Agent-Based Model to Explain Gentrification in European Cities** (MSc thesis, 8/8): Amsterdam · Utrecht · Milan.

EDUCATION

MSc Data Science & Business Analytics, Bocconi University · 107/110

2023 – 2025

Focus: Finance · Econometrics · Statistics · NLP

Thesis: Building an Agent-Based Model to Explain Gentrification in European Cities

BSc Artificial Intelligence, University of Amsterdam · 7.6/10

2019 – 2023

Minor in Linguistics, University of Amsterdam

Erasmus minor, Università di Bologna (UNIBO)

LANGUAGES & SOFT SKILLS

Dutch C2 · English C2 · Italian B2/C1 · German B2 · Spanish A2

Multi-year project management · Client & stakeholder communication · Presenting and defending methodology to technical and senior audiences · Reliability

PROJECTS

LanguageBuddy: AI language tutor

Self-hosted AI language tutor for Dutch, Italian and Spanish: chat or voice-call LLM tutor with SM-2 spaced repetition; every mistake feeds the next day's exercises, with a real-news reader and 6,200+ CEFR-aligned vocabulary entries.

FastAPI · LLM · TTS · SQLite · Docker

Gentrification agent-based model

MSc thesis: an agent-based model of neighbourhood change for Amsterdam, Utrecht and Milan, driven by income, affordability and attractiveness (streetview 'beauty' classification, GTFS connectivity, greenery, neighborhood sentiment). A two-tenure social-housing extension is in progress.

Python/GeoPandas · Large Vision Model · PostGIS · Agent-based Modeling

Connectivity & walkability scoring (KR&A)

Led a saturation-validated connectivity and walkability score at parcel resolution, rolled out across 38 EU/NA/APAC markets from TBs of data and 100s of sources.

PostGIS · S3 · Dijkstra & Network Science · Distributed Compute

Monthly house-price index, 13 EU countries (KR&A)

Web-scraping pipeline across 13 countries feeding a log-price hedonic regression; monthly indices tested as disaggregation indicators for Eurostat quarterly HPis.

Python · Scrapy/Selenium · PostGIS · MongoDB · R

Research pipelines as production systems

Airflow 3 pipelines hosting every personal research project, scaling from a laptop to a multi-machine CeleryExecutor cluster; idempotency guards, custom operators and agentic monitoring.

Airflow · Celery · Docker · CI · DuckDB

SponsoredBye: sponsor-segment detection

Text-only sponsor-skipper for YouTube: sentence-T5 embeddings feed a BiLSTM sequence tagger that maps sponsored sentences back to timestamps, cutting segmentation error from 99% to 16% WindowDiff.

TensorFlow · BiLSTM · sentence-T5 · MongoDB · Huggingface

FishFinder: photo-to-species ID

Flutter app identifying 63 Dutch fish species from a photo, fully on-device: ~3,000 hand-annotated photos masked with Segment Anything, a fine-tuned ResNet50 compressed to an 8.8 MB TFLite model.

Flutter/Dart · TFLite · ResNet50 · Segment Anything Model (SAM) · Firebase

Predicting flooding risk from local features

BSc thesis: explaining flood risk from 33 local features across ~45,000 European locations (100GB+ of data); a Random Forest reaches 97.5% on the binary 20-year flood question.

scikit-learn · Random Forest · raster data · GIS